Formalising the Radon-Nikodym Theorem

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1 Introduction

For this project, we formalise the Radon-Nikodym theorem, one of the most fundamental results in measure theory. The theorem characterises when a measure ν can be expressed as an integral with respect to another measure μ.

Theorem 1 (Radon-Nikodym)**.** *Let μ and ν be σ-finite measures on a measurable space (X, Σ). Then*

$$\nu \ll \mu \quad \Longleftrightarrow \quad \exists f : X \to [0, \infty], \text{ measurable, such that } \nu = f \cdot \mu.$$

Here $\nu \ll \mu$ denotes *absolute continuity*: $\mu(A) = 0$ implies $\nu(A) = 0$ for all measurable A. The function f is called the *Radon-Nikodym derivative* of ν with respect to μ, written $\frac{d\nu}{d\mu}$.

The Radon-Nikodym theorem is foundational throughout mathematics:

- In probability theory, conditional expectation is defined as a Radon-Nikodym derivative.
- In statistics, likelihood ratios are Radon-Nikodym derivatives.
- In functional analysis, it enables the representation of linear functionals.
- In mathematical finance, the Girsanov theorem relies on Radon-Nikodym derivatives for change of measure.

The proof requires a chain of intermediate results:

Hahn Decomposition $\Rightarrow$ Jordan Decomposition $\Rightarrow$ Lebesgue Decomposition $\Rightarrow$ Radon-Nikodym.

Lean is an interactive theorem prover that allows mathematical proofs to be written in a formal language and checked by a computer. In this project, I formalise the Radon-Nikodym theorem in Lean 4 using the Mathlib library, building up through the necessary intermediate results.

2 Mathematical Background

2.1 Signed Measures

A *signed measure* on a measurable space (X, Σ) is a σ-additive function $s : \Sigma \to \mathbb{R}$ with $s(\emptyset) = 0$. Unlike positive measures, signed measures can take negative values.

Definition 1 (Positive and Negative Sets)**.** *Let s be a signed measure on (X, Σ).*

- *A measurable set P is* positive *for s if $s(B) \geq 0$ for all measurable $B \subseteq P$.*
- *A measurable set N is* negative *for s if $s(B) \leq 0$ for all measurable $B \subseteq N$.*

2.2 Hahn Decomposition

Theorem 2 (Hahn Decomposition (Theorem 4.13))**.** *For any signed measure α on $(X, \mathcal{A})$, there exist a positive set A and a negative set B such that $X = A \cup B$ and $A \cap B = \emptyset$.*

Proof sketch. Define $b = \inf\{\alpha(B) : B \text{ is a negative set}\}$. Take negative sets B_n with $\alpha(B_n) \to b$; their union $B = \bigcup B_n$ is negative with $\alpha(B) = b > -\infty$. Setting $A = B^c$, one shows A is positive by contradiction: any subset of A with negative measure would contain a negative subset (Lemma 4.16) that could be adjoined to B, contradicting the minimality of b. □

2.3 Jordan Decomposition

Theorem 3 (Jordan Decomposition (Theorem 4.19))**.** *Let α be a signed measure on $(X, \mathcal{A})$. There exists a unique pair (α_+, α_-) of mutually singular (positive) measures on $(X, \mathcal{A})$ such that*

$$\alpha = \alpha_+ - \alpha_-.$$

Proof sketch. Let (A, B) be a Hahn decomposition for α. Define $\alpha_+(E) = \alpha(E \cap A)$ and $\alpha_-(E) = -\alpha(E \cap B)$. Since A is positive and B is negative, both $\alpha_\pm$ are positive measures, and one checks $\alpha_+ - \alpha_- = \alpha$ and $\alpha_+ \perp \alpha_-$ (with witnessing set A). Uniqueness follows from the essential uniqueness of the Hahn decomposition. □

Definition 2 (Total Variation)**.** *The* total variation measure *of a signed measure α is defined as*

$$|\alpha| = \alpha_+ + \alpha_-.$$

2.4 Absolute Continuity and Mutual Singularity

Definition 3 (Absolute Continuity)**.** *A measure ν is* absolutely continuous *with respect to μ, written $\nu \ll \mu$, if*

$$\mu(A) = 0 \implies \nu(A) = 0 \quad \textit{for all measurable } A.$$

Definition 4 (Mutual Singularity)**.** *Two measures μ and ν are* mutually singular*, written $\mu \perp \nu$, if there exists a measurable set A such that $\mu(A) = 0$ and $\nu(A^c) = 0$.*

Lemma 1. *If $\nu \ll \mu$ and $\nu \perp \mu$, then $\nu = 0$.*

Proof. Let A be the set from mutual singularity with $\mu(A) = 0$ and $\nu(A^c) = 0$. For any measurable B:

$$\nu(B) = \nu(B \cap A) + \nu(B \cap A^c) \leq \nu(A) + \nu(A^c) = 0 + 0 = 0,$$

where $\nu(A) = 0$ follows from $\nu \ll \mu$ and $\mu(A) = 0$. □

2.5 Lebesgue Decomposition

Theorem 4 (Lebesgue Decomposition (Theorem 4.20))**.** *Let μ, ν be two (positive) σ-finite measures on $(X, \mathcal{A})$. Then there exists a unique pair of measures (ν_s, ν_{ac}), the* Lebesgue decomposition *of ν with respect to μ, satisfying:*

$$\nu = \nu_s + \nu_{ac}$$

where $\nu_s \perp \mu$ (singular part) and $\nu_{ac}(A) = \int_A f \, d\mu$ for some measurable $f : X \to [0, \infty]$ (absolutely continuous part).

Proof sketch. By σ-finiteness one may reduce to the case where μ and ν are finite. Introduce the set of "dominated densities" $\mathcal{G} = \{g \geq 0 \text{ measurable} : \int_A g\, d\mu \leq \nu(A)\ \forall A\}$, which is closed under pointwise maxima. Taking a maximising sequence and passing to the limit via monotone convergence yields $f \in \mathcal{G}$ with $\int f\, d\mu = \sup_{g \in \mathcal{G}} \int g\, d\mu$. Setting $\nu_s = \nu - f \cdot \mu$, a Hahn-decomposition argument (Lemma 4.22) shows $\nu_s \perp \mu$, and uniqueness follows from the fact that absolute continuity and mutual singularity together force a measure to be zero. $\square$

3 Mathematical Proof of Radon-Nikodym

Direction 1: Density implies Absolute Continuity

Suppose $\nu = f \cdot \mu$ for some measurable $f \geq 0$, meaning $\nu(A) = \int_A f\, d\mu$ for all measurable A. If $\mu(A) = 0$, then

$$\nu(A) = \int_A f\, d\mu = 0,$$

since the integral over a null set is zero. Hence $\nu \ll \mu$.

Direction 2: Absolute Continuity implies Density

Assume $\nu \ll \mu$ with both measures σ-finite. By the Lebesgue decomposition:

$$\nu = \nu_s + \nu_{ac}$$

where $\nu_s \perp \mu$ and $\nu_{ac} = f \cdot \mu$ for some measurable $f \geq 0$.

Since $\nu \ll \mu$, we have $\nu_s \ll \mu$ (as $\nu_s \leq \nu$). But $\nu_s \perp \mu$ by definition of the Lebesgue decomposition. By the key lemma, $\nu_s \ll \mu$ and $\nu_s \perp \mu$ implies $\nu_s = 0$.

Therefore $\nu = \nu_{ac} = f \cdot \mu$, and $f = \frac{d\nu}{d\mu}$ is the Radon-Nikodym derivative.

4 Implementation Discussion

The mathematical development translates into Lean by working within Mathlib's measure theory library, which provides a rich set of abstractions for the objects involved. The key types and notations are:

- signed measures as `SignedMeasure` α, with positive/negative sets written `0 ≤[P] s` and `s ≤[P^c] 0`,
- absolute continuity written as $\nu \ll \mu$ (i.e. `Measure.AbsolutelyContinuous`),
- mutual singularity written as $\nu \perp_m \mu$ (i.e. `Measure.MutuallySingular`),
- the density construction μ`.withDensity f`, which builds the measure $A \mapsto \int_A f\, d\mu$,
- the Radon-Nikodym derivative ν`.rnDeriv` μ and the singular part ν`.singularPart` μ,
- the Jordan decomposition accessed via `s.toJordanDecomposition`.

The formalisation is split into six parts, mirroring the chain of results in the mathematical development.

Part 1 (Hahn Decomposition). The Hahn decomposition theorem asserts that every signed measure s admits a partition of the space into a positive set P and a negative set P^c. Mathlib

already provides this, so our theorem `hahn_decomposition_exists` is a thin wrapper around the library. The auxiliary result (Lemma 4.16 in the lecture notes) extracts a negative subset from any set of negative measure, again wrapping existing Mathlib machinery.

```
/-- The Hahn Decomposition Theorem: Every signed measure admits a decomposition
X = P ∪ N where P is positive and N is negative. -/
theorem hahn_decomposition_exists :
    ∃ (P : Set α), MeasurableSet P ∧ 0 ≤[P] s ∧ s ≤[Pᶜ] 0 := by
  exact SignedMeasure.exists_compl_positive_negative s

/-- A set of negative measure contains a negative subset (Lemma 4.16). -/
theorem exists_negative_subset_of_negative_measure {i : Set α} (hi_neg : s i < 0) :
    ∃ j ⊆ i, MeasurableSet j ∧ s ≤[j] 0 ∧ s j < 0 := by
  obtain ⟨j, hj_meas, hji, hj_neg, hj_val⟩ :=
    SignedMeasure.exists_subset_restrict_nonpos hi_neg
  exact ⟨j, hji, hj_meas, hj_neg, hj_val⟩
```

Part 2 (Jordan Decomposition). The Jordan decomposition writes a signed measure as $s = s^+ - s^-$ where s^+ and s^- are mutually singular positive measures. In Lean, we define `positivePart` and `negativePart` by extracting the components of `s.toJordanDecomposition`. Mathlib records these as finite measures automatically via type class instances. The decomposition identity `jordan_decomposition_eq` unfolds the definitions and appeals to the library lemma `toSignedMeasure_toJordanDecomposition`, while mutual singularity is inherited directly from the Jordan decomposition structure.

```
/-- The positive part of the Jordan decomposition. -/
noncomputable def positivePart (s : SignedMeasure α) : Measure α :=
  s.toJordanDecomposition.posPart

/-- The negative part of the Jordan decomposition. -/
noncomputable def negativePart (s : SignedMeasure α) : Measure α :=
  s.toJordanDecomposition.negPart

/-- Instance: the positive part is a finite measure. -/
instance (s : SignedMeasure α) : IsFiniteMeasure (positivePart s) :=
  s.toJordanDecomposition.posPart_finite

/-- Instance: the negative part is a finite measure. -/
instance (s : SignedMeasure α) : IsFiniteMeasure (negativePart s) :=
  s.toJordanDecomposition.negPart_finite

/-- The Jordan Decomposition: s = s⁺ - s⁻.
This is Theorem 4.19 in the lecture notes. -/
theorem jordan_decomposition_eq :
    s = (positivePart s).toSignedMeasure - (negativePart s).toSignedMeasure := by
  unfold positivePart negativePart
  exact s.toSignedMeasure_toJordanDecomposition.symm

/-- The positive and negative parts are mutually singular. -/
```

```
theorem jordan_parts_mutuallySingular :
    positivePart s ⊥ₘ negativePart s := by
  unfold positivePart negativePart
  exact s.toJordanDecomposition.mutuallySingular

/-- The total variation measure |s| = s⁺ + s⁻. -/
noncomputable def totalVariation (s : SignedMeasure α) : Measure α :=
  positivePart s + negativePart s
```

Part 3 (Absolute Continuity and Mutual Singularity). The two fundamental relations between measures—absolute continuity ($\nu \ll \mu$) and mutual singularity ($\nu \perp_m \mu$)—are "opposite" in a precise sense: if both hold simultaneously, then $\nu = 0$. This key lemma is the engine that drives the Radon-Nikodym theorem. We first record several basic facts: that absolute continuity is definitionally equivalent to preserving null sets, that `μ.withDensity f ≪ μ` (the easy direction of Radon-Nikodym), that mutual singularity is symmetric, and that absolute continuity is transitive.

```
/-- Absolute continuity: ν ≪ μ means every μ-null set is ν-null. -/
lemma ac_def : ν ≪ μ ↔ ∀ s, μ s = 0 → ν s = 0 :=
  Iff.rfl

/-- If ν = f • μ (ν has density f w.r.t. μ), then ν ≪ μ. -/
theorem absolutelyContinuous_withDensity (f : α → ℝ≥0∞) :
    μ.withDensity f ≪ μ :=
  withDensity_absolutelyContinuous μ f

/-- Mutual singularity is symmetric. -/
theorem mutuallySingular_symm : μ ⊥ₘ ν ↔ ν ⊥ₘ μ :=
  ⟨Measure.MutuallySingular.symm, Measure.MutuallySingular.symm⟩

/-- Transitivity of absolute continuity. -/
theorem absolutelyContinuous_trans {ρ : Measure α}
    (hνμ : ν ≪ μ) (hμρ : μ ≪ ρ) : ν ≪ ρ :=
  hνμ.trans hμρ
```

Key lemma ($\nu \ll \mu$ and $\nu \perp_m \mu$ implies $\nu = 0$). The proof of `eq_zero_of_ac_of_mutuallySingular` is the most substantial manual argument in the formalisation. From mutual singularity we obtain a separating set A with $\nu(A) = 0$ and $\mu(A^c) = 0$. To show $\nu(s) = 0$ for an arbitrary measurable set s, we split it as $s = (s \cap A) \cup (s \cap A^c)$. The first piece satisfies $\nu(s \cap A) \leq \nu(A) = 0$; the second satisfies $\mu(s \cap A^c) \leq \mu(A^c) = 0$, and absolute continuity then gives $\nu(s \cap A^c) = 0$. The `calc` block assembles these estimates.

```
/-- Key lemma: If ν ≪ μ and ν ⊥ μ, then ν = 0.
This is crucial for the Radon-Nikodym theorem. -/
theorem eq_zero_of_ac_of_mutuallySingular (hac : ν ≪ μ) (hms : ν ⊥ₘ μ) :
    ν = 0 := by
  -- From mutual singularity, get the separating set A
  rcases hms with ⟨A, hA_meas, hνA, hμAc⟩
  -- Show ν = 0 by showing ν(s) = 0 for all measurable s
```

```
  ext s hs
  -- Split s into s ∩ A and s ∩ Aᶜ
  have hsplit : s = (s ∩ A) ∪ (s ∩ Aᶜ) := by rw [inter_union_compl]
  have hdisj : Disjoint (s ∩ A) (s ∩ Aᶜ) := by
    rw [Set.disjoint_iff]
    intro x ⟨⟨_, hxA⟩, ⟨_, hxAc⟩⟩
    exact hxAc hxA
  calc ν s = ν ((s ∩ A) ∪ (s ∩ Aᶜ)) := by rw [← hsplit]
    _ = ν (s ∩ A) + ν (s ∩ Aᶜ) := measure_union hdisj (hs.inter hA_meas.compl)
    _ = 0 + 0 := by
        congr 1
        -- ν(s ∩ A) ≤ ν(A) = 0
        · exact le_antisymm
            (measure_mono inter_subset_right |>.trans (le_of_eq hνA)) (zero_le _)
        -- ν(s ∩ Aᶜ) = 0 because μ(s ∩ Aᶜ) ≤ μ(Aᶜ) = 0 and ν ≪ μ
        · have hμ : μ (s ∩ Aᶜ) ≤ μ Aᶜ := measure_mono inter_subset_right
          exact hac (hμ.trans (le_of_eq hμAc) |>.antisymm (zero_le _))
    _ = 0 := add_zero 0
```

Part 4 (Lebesgue Decomposition). The Lebesgue decomposition states that every σ-finite measure ν can be written as $\nu = \nu_s + f \cdot \mu$ where $\nu_s \perp \mu$ and $f = d\nu/d\mu$ is the Radon-Nikodym derivative. Mathlib provides the heavy machinery for this through its `HaveLebesgueDecomposition` type class, which is automatically synthesised for σ-finite measures. We wrap the four key facts: the decomposition identity, mutual singularity of the singular part, measurability of the Radon-Nikodym derivative, and absolute continuity of the density part.

```
/-- The Lebesgue Decomposition: ν = νₛ + (dν/dμ) • μ where νₛ ⊥ μ.
This is Theorem 4.20 in the lecture notes. -/
theorem lebesgue_decomposition :
    ν = ν.singularPart μ + μ.withDensity (ν.rnDeriv μ) :=
  ν.haveLebesgueDecomposition_add μ

/-- The singular part is mutually singular to μ. -/
theorem singularPart_mutuallySingular' (μ ν : Measure α) :
    ν.singularPart μ ⊥ₘ μ :=
  ν.mutuallySingular_singularPart μ

/-- The Radon-Nikodym derivative is measurable. -/
theorem rnDeriv_measurable' (μ ν : Measure α) : Measurable (ν.rnDeriv μ) :=
  Measure.measurable_rnDeriv ν μ

/-- The absolutely continuous part is indeed absolutely continuous. -/
theorem withDensity_rnDeriv_ac (μ ν : Measure α) :
    μ.withDensity (ν.rnDeriv μ) ≪ μ :=
  withDensity_absolutelyContinuous μ (ν.rnDeriv μ)
```

Part 5 (Radon-Nikodym Theorem). The main theorem `radon_nikodym` combines all the preceding results. The reverse direction ($\Leftarrow$) is straightforward: if $\nu = f \cdot \mu$, then absolute continuity follows from the library. The forward direction ($\Rightarrow$) is the heart of the argument. We use

ν.rnDeriv μ as the candidate density and invoke the Lebesgue decomposition to write $\nu = \nu_s + f \cdot \mu$. To eliminate the singular part, we show $\nu_s \ll \mu$ by proving $\nu_s(s) \leq \nu(s)$ via the decomposition and then appealing to $\nu \ll \mu$. Combined with $\nu_s \perp_m \mu$ from the Lebesgue decomposition, the key lemma gives $\nu_s = 0$, and a `calc` block concludes $\nu = f \cdot \mu$.

```
/-- The Radon-Nikodym Theorem: ν ≪ μ iff ν has a density with respect to μ.
This is Corollary 4.23 in the lecture notes. -/
theorem radon_nikodym :
    ν ≪ μ ↔ ∃ f : α → ℝ≥0∞, Measurable f ∧ ν = μ.withDensity f := by
  constructor
  · -- Direction (⇒): Absolute continuity implies existence of density
    intro hac
    -- The Radon-Nikodym derivative provides the density
    use ν.rnDeriv μ
    constructor
    · exact Measure.measurable_rnDeriv ν μ
    · -- By Lebesgue decomposition: ν = νₛ + f • μ
      have hdecomp := lebesgue_decomposition (μ := μ) (ν := ν)
      -- νₛ ⊥ μ
      have hms := singularPart_mutuallySingular' (μ := μ) (ν := ν)
      -- νₛ ≪ μ because νₛ ≤ ν and ν ≪ μ
      have hac_singular : ν.singularPart μ ≪ μ := by
        intro s hs
        have h : ν.singularPart μ s ≤ ν s := by
          calc ν.singularPart μ s
              ≤ ν.singularPart μ s + μ.withDensity (ν.rnDeriv μ) s :=
                le_add_of_nonneg_right (zero_le _)
            _ = (ν.singularPart μ + μ.withDensity (ν.rnDeriv μ)) s :=
                (Measure.add_apply _ _ _).symm
            _ = ν s := by rw [← hdecomp]
        have hν : ν s = 0 := hac hs
        exact le_antisymm (h.trans (le_of_eq hν)) (zero_le _)
      -- Since νₛ ≪ μ and νₛ ⊥ μ, we have νₛ = 0
      have hzero := eq_zero_of_ac_of_mutuallySingular hac_singular hms
      -- Therefore ν = f • μ
      calc ν = ν.singularPart μ + μ.withDensity (ν.rnDeriv μ) := hdecomp
        _ = 0 + μ.withDensity (ν.rnDeriv μ) := by rw [hzero]
        _ = μ.withDensity (ν.rnDeriv μ) := zero_add _
  · -- Direction (⇐): Existence of density implies absolute continuity
    rintro ⟨f, _, hf_eq⟩
    rw [hf_eq]
    exact withDensity_absolutelyContinuous μ f
```

Corollaries. Two important consequences follow immediately. The first, `withDensity_rnDeriv_eq'`, states that if $\nu \ll \mu$ then ν equals the measure built from its own Radon-Nikodym derivative. The second, `rnDeriv_unique`, establishes that the derivative is unique μ-almost everywhere: if two measurable functions f and g define the same measure via `withDensity`, they must agree μ-a.e., using Mathlib's `withDensity_eq_iff_of_sigmaFinite`.

```
/-- Corollary: If ν ≪ μ, then μ.withDensity (ν.rnDeriv μ) = ν. -/
theorem withDensity_rnDeriv_eq' (hac : ν ≪ μ) :
    μ.withDensity (ν.rnDeriv μ) = ν :=
  Measure.withDensity_rnDeriv_eq ν μ hac

/-- The Radon-Nikodym derivative is unique almost everywhere. -/
theorem rnDeriv_unique (μ : Measure α) [SigmaFinite μ] {f g : α → ℝ≥0∞}
    (hf : Measurable f) (hg : Measurable g)
    (hf_eq : μ.withDensity f = μ.withDensity g) :
    f =ᵐ[μ] g := by
  exact (withDensity_eq_iff_of_sigmaFinite hf.aemeasurable hg.aemeasurable).mp hf_eq
```

Part 6 (Applications). Finally, we record two standard consequences of the Radon-Nikodym theorem. The *chain rule* states that for $\nu \ll \mu$, the Radon-Nikodym derivatives compose multiplicatively: $\frac{d\nu}{d\mu} \cdot \frac{d\mu}{d\rho} = \frac{d\nu}{d\rho}$ holds ρ-almost everywhere. The *self-derivative* states that $\frac{d\nu}{d\nu} = 1$ a.e. Both are direct applications of Mathlib library lemmas.

```
/-- Chain rule for Radon-Nikodym derivatives.
If ν ≪ μ, then ν.rnDeriv μ * μ.rnDeriv ρ =ᵐ[ρ] ν.rnDeriv ρ. -/
theorem rnDeriv_chain (hνμ : ν ≪ μ) :
    ν.rnDeriv μ * μ.rnDeriv ρ =ᵐ[ρ] ν.rnDeriv ρ :=
  Measure.rnDeriv_mul_rnDeriv hνμ

/-- Self-derivative: ν.rnDeriv ν = 1 a.e. -/
theorem rnDeriv_self : ν.rnDeriv ν =ᵐ[ν] 1 :=
  Measure.rnDeriv_self ν
```

5 Challenges

The primary challenge in this project was managing the complexity of the theorem's dependencies. Unlike simpler results that can be proved in isolation, the Radon-Nikodym theorem requires a carefully constructed tower of results: Hahn decomposition, Jordan decomposition, and Lebesgue decomposition must all be in place before the main theorem can be stated and proved.

Working with Mathlib's measure theory library presented both advantages and challenges. On one hand, the library provides sophisticated machinery for signed measures, absolute continuity, and mutual singularity. On the other hand, finding the right lemmas and understanding their precise statements required significant exploration. For instance, the relationship between `SignedMeasure`, `JordanDecomposition`, and `Measure` involves several coercion and conversion functions that had to be understood.

Another challenge was the handling of σ-finiteness assumptions. While the mathematical statement of Radon-Nikodym requires σ-finite measures, Lean requires these assumptions to be explicitly tracked through type class instances like `[SigmaFinite μ]`. Ensuring these instances were available in the right places required careful attention.

6 Reflection

Formalising the Radon-Nikodym theorem revealed the elegant logical structure underlying this fundamental result. The proof strategy—using the Lebesgue decomposition and the key lemma

that absolute continuity plus mutual singularity implies zero—is conceptually simple once stated clearly, but its formalisation requires precise bookkeeping of measure-theoretic concepts.

The experience highlighted how measure theory in Mathlib is designed around a few key abstractions. The notions of `AbsolutelyContinuous` ($\ll$) and `MutuallySingular` ($\perp_m$) are defined in terms of null sets, and most theorems about measures can be reduced to arguments about these two relations. The Radon-Nikodym theorem is essentially a statement about when absolute continuity can be "witnessed" by a density function.

The project also demonstrated the power of building on existing library results. While I stated and proved the key structural theorems (like the Jordan decomposition and the main Radon-Nikodym equivalence), the underlying technical machinery—such as the existence of the Lebesgue decomposition and the measurability of the Radon-Nikodym derivative—was provided by Mathlib. This allowed the formalisation to focus on the conceptual content of the theorem rather than low-level measure-theoretic details.

Acknowledgements

I would like to thank Cursor, an AI-powered code editor powered by large language models from OpenAI, Google, and Anthropic, for its assistance during the development of this project. Its suggestions and explanations were helpful in navigating the Mathlib library and debugging Lean code.

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